| Qns | Answer | Marks |
|---|---|---|
| 5(a) | $R_{diode} = \frac{V}{I} = \frac{0.56}{6.0 \times 10^{-3}} = 93\ \Omega$ | A1 |
| 5(b) | $E = V_{diode} + V_{R3}$<br>$1.2 = 0.56 + (6.0 \times 10^{-3})R_3$<br>$R_3 = 110\ \Omega$ | C1<br>A1 |
| 5(c) | $I = \frac{E}{R_1 + R_2}$<br>$= \frac{1.2}{50 + 200}$<br>$= 4.8 \times 10^{-3}$ A | C1<br>A1 |
| 5(d) | $I_{total} = 6.0 \times 10^{-3} + 4.8 \times 10^{-3} = 1.1 \times 10^{-2}$ A<br>$P = I_{total}E = (1.1 \times 10^{-2})(1.2)$<br>$P = 1.3 \times 10^{-2}$ W | C1<br>A1 |
| 5(e) | $\frac{x}{1.0} = \frac{50}{250}$<br>$x = 0.20$ m | C1<br>A1 |

| Qns | Answer | Marks |
|---|---|---|
| **6(a)(i)** | The magnetic force is downwards (by FLHR, *B* points into the paper and current is to the left) in the region between the plates, hence the electric force must be upwards.<br><br>Since the particles are negatively charged, the electric field must be downwards (opposite to the electric force). Hence, P is positive.<br><br>(Plate Q is grounded, hence is at 0 V.) | **B1**<br><br><br><br>**B1** |
| **6(a)(ii)** | To move undeflected, the net force must be zero. | |